Lesson plan · for the teacher

Loops: for, range and while

Programming · Level 2 · Lesson 3

Time	Materials
40 min	Board and chalk; Number cards 1 to 10; Your phone with this lesson downloaded

Learning goals

- Repeat lines with for and range.
- Say exactly where range stops.
- Build a total in a loop, and use while.

Activity: Be the loop

1. Warm-up (5 min): Clap a rhythm and ask pupils to repeat it five times. Write it as for i in range(5): clap(). Ask how to make it ten times.
2. Teach (10 min): Write for n in range(1, 11): print(7, "x", n, "=", 7 * n). Read it aloud line by line; show that range stops before 11.
3. Be the loop (10 min): One pupil is n, holding number cards 1 to 5; another is total, with chalk. Each turn, n shows the next card and total adds it on the board. The class checks: 15.
4. Predict and run (10 min): Change range(1, 6) to range(1, 11), then range(0, 5). Pairs predict the total each time; you run it on your phone.
5. Wrap-up (5 min): Ask why range(1, 10) misses 10, and where else that could cause a bug. Then give the exit ticket.

No devices are needed until “Predict and run”: number cards and chalk are enough, and one phone runs the programs.

Discussion

- Why does range stop before its second number?
- When would you use while instead of for?
- Which job in our school repeats, and could be a loop?

Exit ticket

- Write a loop that prints 2, 4, 6, 8, 10, and say which range you used.

Try it at school tomorrow
Write for n in range(1, 6): print(n * 7) on the board. Ask pupils to predict every line, then how to make it go up to 10. Run each version on your phone.

Practice questions · for you

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1 **This loop prints one line for each value. How many lines does it print?**

```
for i in range(0, 5):
    print("Line", i)
```

○ 4 ○ 5 ○ 6

2 **Finish the program: it asks which times table to show, then prints it from 1 to 10, like 7 x 1 = 7.**

```
n = int(input("Which table? "))
# print n x 1 = ... up to n x 10
```

Write your program:

3 **Fill the gaps so the program adds up the numbers 1 to 10 and prints 55.**

```
total = 0
for n in _____(1, _____):
    total = total + n
print(total)
```

Word bank: `11` `list` `range` `10`

4 **This program should print a line for each of the 30 pupils on the register, but the last pupil is missing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  # Print the register numbers 1 to 30
2  for n in range(1, 30):
3      print("Pupil", n)
```

5 **Match each loop to how many times it repeats.**

for i in range(5):	• • • •	once for each mark in the list
for m in marks:	• • • •	exactly 5 times
while total < 100:	• • • •	for ever, unless something stops it
while True:	• • • •	until total reaches 100

✎ **Which job do you repeat every week (marks to add up, a register to count, a table to write)? Write the loop that would do it: what repeats, and how many times?**

Check your answers

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1 **5**

range(0, 5) gives 0, 1, 2, 3 and 4: five values, so five lines.

2

```
n = int(input("Which table? "))
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

One for loop replaces ten print lines. range(1, 11) gives 1 to 10, and n * i works out each answer, so the same program writes any table.

3

```
total = 0
for n in range(1, 11):
    total = total + n
print(total)
```

range(1, 11) gives 1 to 10, because range stops before its second number.

4 **Line 2:** `for n in range(1, 31):`

range(1, 30) stops before 30, so it gives 1 to 29: off by one.

5

for i in range(5): — exactly 5 times
for m in marks: — once for each mark in the list
while total < 100: — until total reaches 100
while True: — for ever, unless something stops it

for repeats a known number of times: once per value.